

Lorenzo Corti

Residence: Netherlands – Nationality: Italian

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Summary

I work on **tools** and **methods** for diagnosing large **vision and language models**. My work sits at the intersection of **NLP** and **interpretable ML**, where I develop rigorous **evaluations** and integrate modern vision and language models into end-to-end **applications**. I have hands-on experience with frameworks such as **PyTorch**, **Hugging Face Transformers**, and **spaCy**, and I am particularly interested in building reliable and **robust multimodal systems** and workflows.

I am currently pursuing a PhD at Delft University of Technology. Previously, I received my MSc and BSc in Computer Science and Engineering from Politecnico di Milano.

Skills

Languages Python, Java, C, Javascript
Data Management Relational DBs and NoSQL
Web Development Spring, FastAPI, Django
DevOps Docker, VPS, Git

Machine Learning NumPy, Pandas, scikit-learn, spaCy
Deep Learning Pytorch, Captum, Huggingface
Transformers, Datasets, and Evaluate
Project Management Research design and planning

Education

Delft University of Technology, Delft, NL

Feb 2022 – 2027*

PhD Candidate in Computer Science

**Finalising my dissertation and can terminate earlier than Feb. 2027.*

- *Dissertation:* Beyond Benchmarks – Diagnosis of Vision and Language Model Behaviours
- *Supervision:* Prof.Dr.Ir. Geert-Jan Houben (promotor) and Dr. Jie Yang (copromotor)

Politecnico di Milano, Milan, IT

Master of Science in Computer Science and Engineering (110/110)

Sept 2017 – Dec 2019

- *Thesis:* Time Conditional Generative Adversarial Networks for Augmentation of Irregularly Sampled Time Series
- *Supervision:* Marco Brambilla (Politecnico di Milano) and Pavlos Protopapas (Harvard University)

Bachelor of Science in Computer Science and Engineering (98/110)

Sept 2014 – July 2017

- *Thesis Project:* Development of a customer relationship management system for an advertising company.
- *Deliverables:* Requirement analysis with i* framework, UML design, and documented codebase.

Professional Experience

Doctoral Candidate

Feb 2022 – 2027*

Delft University of Technology, Delft, NL

- **End-to-End Diagnosis of Vision Language Models:** Method to diagnose the behaviours of VLMs, verify observable ones, and quantify concept-level effects. Evaluated on four VLMs across four benchmarks for multiple-choice and open-ended visual question answering tasks. Concept-level effects are quantified using a Double-ML causal inference estimator, fit onto synthetically occluded visual inputs.
‣ *Implementation:* Hugging Face Transformers, PyTorch, and NumPy to self-host and work with VLMs, Hugging Face Datasets, Pandas, OpenCV, SentenceTransformers, spaCy, and scikit-learn for data preprocessing and analysis, IETrans and NetworkX for scene graph extraction and analysis, DoWhy and EconML for causal analysis, and Matplotlib and Seaborn for visualisation.
- **Toolkit for LLM Behaviour Diagnosis:** Design and development of a toolkit to diagnose LLM behaviours on known linguistic structures. The toolkit supports encoder-only, encoder-decoder, and decoder-only transformer-based LLMs, different NLP datasets, known NLP metrics, inspection and tagging of LLM outputs, and behavioural resting and reporting. The tool is externally validated with NLP industry practitioners.
‣ *Implementation:* Hugging Face Transformers and Datasets, PyTorch, spaCy, and NumPy as foundation, Pandas and scikit-learn for metrics computation, Captum for interpretable ML support, FastAPI and MongoDB for processing backend, Django and MySQL for frontend, and Docker for containerisation and self-hosting.
- **Extraction of Behaviours of Vision Models:** Explainable AI method to extract observable behaviours of CNN-based vision models. Applied the method to vision models in the presence of known biases (introduced via fine-tuning) and evaluated in terms of fidelity, coherence, and accuracy of the behaviours extracted.
‣ *Implementation:* DoWhy for causal estimates, CDT and NetworkX for causal graphs discovery and analysis, NumPy, Pandas, scikit-learn, Matplotlib for data analysis and visualisation.
- Additionally, under the TU Delft AI Lab Design@Scale, I have mentored 8 M.Sc. theses, 5 B.Sc. theses, and 10 student group projects on NLP, language models, model interpretability, human computation, and information systems.

- **GANs for Time Series Imputation:** Continued training from scratch of GANs combining LSTM and Convolutional networks. Implemented different training regimes and normalisations to prevent mode collapse.
 </> Implementation: TensorFlow, NumPy, scikit-learn.
- **IDiOM:** Modular and open-source NLP data analysis pipeline for disinformation. Implements data collection from social and news media sources and runs natural language processing and machine learning analyses.
 </> Implementation: NumPy, scikit-learn, Hugging Face Transformers, spaCy, MongoDB, and custom data ingestion modules.
- **VaccinItaly:** Analysis of the Italian discourse on COVID-19 (perception and vaccine intake/hesitancy) on social media (Twitter and Facebook) and news outlets. Includes temporal, geographical and lexical analyses as well as training of stance classifiers (pro- or anti-vaccination).
 </> Implementation: IDiOM pipeline (above) and self-developed code with NumPy, scikit-learn, spaCy, and Matplotlib.
- **COCTEAU:** Gamified, crowdsourcing platform aiming to enable large-scale citizen engagement.
 </> Implementation: Spring Boot, Java, jQuery, Bootstrap, and MySQL.
- **PERSEUS:** Document search and analysis platform leveraging language modelling for NER, topic extraction, and document analysis.
 </> Implementation: Python FastAPI, Hugging Face Transformers, and scikit-learn backend, jQuery and Bootstrap frontend, and Elasticsearch storage.

Languages

- **Italian**, Native.
- **English**, Full Professional Proficiency.
- **Spanish**, Elementary.

Selected Publications

1. *End-to-End Diagnosis of Vision Language Models*. **Lorenzo Corti**, Jie Yang. *Under review*.
2. *Toolkit for LLM Behaviour Diagnosis*. **Lorenzo Corti**, Sebastiaan Beekman, Jie Yang. *Under review*.
3. *A.I. Robustness: a Human-Centered Perspective on Technological Challenges and Opportunities*. Andrea Tocchetti*, **Lorenzo Corti***, Agathe Balayn*, Mireia Yurrita, Philip Lippmann, Marco Brambilla, Jie Yang (* equal contribution). In: ACM Computing Surveys. 2025. DOI: <https://doi.org/10.1145/3665926>.
4. *CHIME: Causal Human-in-the-Loop Model Explanations*. Shreyan Biswas, **Lorenzo Corti**, Stefan Buijsman, Jie Yang. In: Proceedings of the AAAI Conference on Human Computation and Crowdsourcing. 2022. DOI: <https://doi.org/10.1609/hcomp.v10i1.21985>.
5. *A Web-Based Co-Creation and User Engagement Method and Platform*. Andrea Tocchetti, **Lorenzo Corti**, Marco Brambilla, Diletta Di Marco. In: Proceedings of International Conference on Web Engineering. 2021. DOI: https://doi.org/10.1007/978-3-030-74296-6_38.

Workshop and Demo Papers

1. *ARTIST: ARTificial Intelligence for Simplified Text*. **Lorenzo Corti**, Jie Yang. In: Generative AI and HCI workshop at CHI. 2023. DOI: <https://doi.org/10.48550/arXiv.2308.13458>.
2. *A content-based approach for the analysis and classification of vaccine-related stances on Twitter: the Italian scenario*. Marco Di Giovanni, **Lorenzo Corti**, Silvio Pavanetto, Francesco Pierri, Andrea Tocchetti, Marco Brambilla. In: Workshop Proceedings of the 15th International AAAI Conference on Web and Social Media. 2021. DOI: <https://doi.org/10.36190/2021.52>.